TECHNICAL REPORT

Assessment of the Stability of the QGEO-KZ GNSS Network

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Executive Summary

This report presents an initial assessment of the temporal stability of the QGEO-KZ GNSS stations. The main objective is to identify the stations showing the most promising behaviour for more detailed evaluation as potential candidates for submission to the International GNSS Service (IGS).

Daily coordinate solutions were obtained with GipsyX and the resulting North, East and Up coordinate time series were analysed with Hector. The analysis considers data continuity, coordinate scatter, offsets, anomalous intervals, estimated velocities and consistency with the regional velocity field.

The initial network list contains 87 stations. A minimum observation span of 2.5 years was adopted and the position and velocity solutions were referred to 1 January 2025. Most of the QGEO series cover approximately three to four years, principally from June 2022 to July 2026.

Horizontal velocities relative to the Eurasian plate were calculated to provide an initial assessment of network consistency and to identify anomalous station behaviour. The resulting field can also support a future evaluation of long-term internal deformation, but the present solution should not yet be interpreted as a definitive deformation model.

More robust results can be obtained after identified processing discontinuities are investigated and corrected and as longer coordinate time series become available. This report addresses coordinate stability only. Other technical and operational requirements would need to be evaluated separately before any station is proposed to the IGS.

1. Introduction

1.1 Background

Within the technical collaboration with QGEO, an initial analysis of the QGEO-KZ GNSS network was performed to evaluate the temporal behaviour of its stations. This is the first systematic computation of daily coordinate time series and station velocities prepared within this collaboration.

The purpose is to identify the stations with the most promising coordinate behaviour for possible future consideration as IGS stations. The present analysis is an initial screening and does not constitute a final selection or an application to the IGS.

1.2 Objectives

The analysis evaluates the availability and continuity of the observations, the stability of the North, East and Up coordinate components, the reliability of preliminary velocity estimates, and the consistency of horizontal velocities after removal of the predicted Eurasian plate motion.

Together, these indicators provide a first basis for distinguishing the most promising stations from stations requiring additional investigation before any further IGS-related assessment.

1.3 Scope and limitations

The current evaluation is based on coordinate time-series behaviour and the preliminary velocity field. Most station records span only three to four years, so the velocity estimates remain sensitive to offsets, anomalous periods and processing discontinuities. The results will become more robust as the observation span increases and the identified discontinuities are resolved.

2. QGEO-KZ Data Set

2.1 Network overview

Table 1 – General information about the QGEO Analysis

Initial network list	87 stations
Minimum time span	2.5 years
Stations meeting the span criterion	85 in the Hector analysis
Reference epoch	1 January 2025
Typical QGEO observation period	June 2022 to July 2026

PARK and EGAN were excluded from the QGEO-specific interpretation because their four-character identifiers duplicate existing station names. Station identifiers must be unique before any future exchange or submission of station products.

2.2 Data availability

For each station, the processing inventory records the first and last observation dates, total time span, number of available files, number of successfully processed files and number of unsuccessful or unavailable solutions. Most stations provide sufficiently dense daily series for an initial stability assessment, although some contain significant gaps or periods of reduced availability.

The station-by-station observation and processing availability for all 85 stations meeting the 2.5-year span criterion is listed below. The initial network list contains 87 stations.

Table 2 – Observation and processing availability for the 85 stations entering the initial Hector analysis

Site	First obs.	Last obs.	Span (yr)	Available	Processed	Not estimated	No solution
AALA	28-06-2022	11-07-2026	4.0	1382	1368	14	0
ATAL	31-08-2022	11-07-2026	3.9	1341	1304	37	0
BBAL	04-01-2023	11-07-2026	3.5	975	949	7	0
BKEG	27-06-2022	11-07-2026	4.0	1075	1065	10	0
BSAR	04-01-2023	10-07-2026	3.5	870	860	7	0
BUSH	27-06-2022	11-07-2026	4.0	1305	1293	12	0
BZHK	27-06-2022	11-07-2026	4.0	1224	1212	12	0
CATB	28-06-2022	11-07-2026	4.0	1160	1142	11	1
CBST	28-06-2022	11-07-2026	4.0	1233	1217	15	0
CERM	28-06-2022	11-07-2026	4.0	1172	1160	11	0
CKSH	28-06-2022	21-06-2026	4.0	1278	1266	12	0
DAKT	27-06-2022	11-07-2026	4.0	1202	1192	10	0
DBOZ	28-06-2022	11-07-2026	4.0	1423	1411	12	0
DEMB	28-06-2022	11-07-2026	4.0	1237	1172	65	0
DKOM	22-09-2022	11-07-2026	3.8	1033	1020	9	0
DSHL	28-06-2022	11-07-2026	4.0	1321	1304	17	0
DYRG	27-06-2022	11-07-2026	4.0	1423	1402	20	0
DZHN	03-08-2022	11-07-2026	3.9	1326	1311	14	0
DZHR	09-09-2022	11-07-2026	3.8	1259	1242	15	0
EATR	28-06-2022	11-07-2026	4.0	1345	1332	12	0
EIND	28-06-2022	11-07-2026	4.0	1404	1381	23	0
EKRB	29-06-2022	11-07-2026	4.0	1348	1328	18	0
EKUL	27-06-2022	06-07-2026	4.0	992	981	10	0
FAJG	27-06-2022	11-07-2026	4.0	1229	1205	12	1
FAKS	27-06-2022	05-07-2026	4.0	1117	1101	11	0
FCKP	18-10-2022	28-06-2026	3.7	1125	1114	10	0
FKRA	01-07-2022	11-07-2026	4.0	1345	1333	12	0
FKTK	25-07-2022	11-07-2026	4.0	696	687	9	0
FOSK	06-02-2023	29-06-2026	3.4	829	822	5	0
FSEM	27-06-2022	29-06-2026	4.0	1179	1167	12	0
FZAY	28-06-2022	11-07-2026	4.0	1077	1067	10	0
HMYA	12-09-2022	15-03-2026	3.5	1124	1110	11	0
HSHO	28-06-2022	11-07-2026	4.0	1352	1331	20	0
HTAR	09-08-2022	24-05-2026	3.8	1022	1012	7	0
HULB	28-06-2022	11-07-2026	4.0	1227	1202	20	0
HZHN	14-03-2023	15-04-2026	3.1	839	835	3	0
LAKS	28-06-2022	11-07-2026	4.0	1190	1177	12	0
LCHP	08-08-2022	11-07-2026	3.9	1262	1250	11	0
LKRT	15-08-2022	11-07-2026	3.9	1253	1242	11	0
LKZT	12-08-2022	11-07-2026	3.9	1017	996	16	0
LORA	28-06-2022	11-07-2026	4.0	1419	1386	33	0
LSAY	29-09-2022	11-07-2026	3.8	1155	1120	24	0

Site	First obs.	Last obs.	Span (yr)	Available	Processed	Not estimated	No solution
LZHK	15-08-2022	11-07-2026	3.9	1192	1162	26	0
MAKD	01-09-2022	21-06-2026	3.8	1169	1156	10	0
MAKT	28-06-2022	11-07-2026	4.0	1355	1338	12	0
MBRS	28-06-2022	11-07-2026	4.0	1411	1396	12	0
MKKL	01-07-2022	11-07-2026	4.0	1291	1279	12	0
MKRG	27-06-2022	11-07-2026	4.0	1259	1244	12	1
MKRJ	28-06-2022	11-07-2026	4.0	1434	1422	12	0
MKSK	28-06-2022	11-07-2026	4.0	1156	1138	15	0
MSAS	28-06-2022	11-07-2026	4.0	1334	1315	19	0
MSAT	28-06-2022	11-07-2026	4.0	1408	1384	24	0
MSYA	30-06-2022	11-07-2026	4.0	1284	1259	25	0
NARS	12-07-2022	11-07-2026	4.0	1165	1154	8	0
NKZR	04-01-2023	11-07-2026	3.5	781	770	4	0
NQZL	27-08-2022	11-07-2026	3.9	970	959	10	0
NZHG	25-07-2022	01-07-2026	3.9	1161	1135	23	0
NZHS	28-06-2022	11-07-2026	4.0	880	579	13	0
PKOS	27-06-2022	29-03-2026	3.8	1222	1210	11	0
PKRB	28-06-2022	11-07-2026	4.0	1353	1341	12	0
PKRM	28-06-2022	11-07-2026	4.0	1034	933	33	11
PTRG	29-06-2022	11-07-2026	4.0	971	956	12	0
PUZK	28-06-2022	11-07-2026	4.0	1359	1342	17	0
PZHT	07-12-2022	11-07-2026	3.6	1135	1130	5	0
RAKT	28-06-2022	11-07-2026	4.0	1317	1303	14	0
RBEY	28-06-2022	21-06-2026	4.0	1263	1015	248	0
RBLS	28-06-2022	11-07-2026	4.0	1117	1104	12	0
RJNO	01-07-2022	11-07-2026	4.0	1108	1094	5	1
RKYZ	04-01-2023	11-07-2026	3.5	996	991	3	0
RTSH	28-06-2022	10-07-2026	4.0	869	849	10	0
SBYA	12-07-2022	11-07-2026	4.0	1015	1006	7	0
SEKB	27-06-2022	21-06-2026	4.0	1300	1288	12	0
SJLZ	22-08-2022	11-07-2026	3.9	1038	1027	11	0
SPAV	05-09-2022	11-07-2026	3.8	954	936	5	0
SSHB	17-11-2022	11-07-2026	3.6	1174	1168	5	0
TNIS	28-06-2022	11-07-2026	4.0	1380	1371	9	0
TPET	04-01-2023	11-07-2026	3.5	951	942	7	0
TTLS	28-06-2022	11-07-2026	4.0	1168	1158	10	0
XKZE	28-06-2022	10-07-2026	4.0	1041	1028	9	0
XSHM	28-06-2022	11-07-2026	4.0	1231	1217	11	0
XTUR	28-06-2022	11-07-2026	4.0	1412	1401	11	0
XZHT	28-06-2022	11-07-2026	4.0	1315	1287	19	0
ZNUR	28-06-2022	11-07-2026	4.0	1375	1364	11	0

3. Processing with GipsyX

3.1 Daily coordinate processing

The available daily RINEX observation files were processed independently for each station with GipsyX/RTGx using the Precise Point Positioning (PPP) strategy. The principal elements of the processing are summarised below.

Table 3 – Summary of GipsyX processing

Software	GipsyX/RTGx
Processing strategy	Independent 24-hour static PPP solution for each station
Precise products	JPL final GNSS orbit and clock products and associated models
Reference-frame mapping	Daily non-fiducial solutions mapped using the seven-parameter transformation
Daily output	One Cartesian position per successfully processed station and day, subsequently expressed as local North, East and Up components

4. Time-Series Analysis with Hector

4.1 Analysis approach

Hector was used to estimate the long-term coordinate trend and its uncertainty from the daily North, East and Up time series. The reference epoch was set to 1 January 2025 and only stations with at least 2.5 years of observations entered the initial network analysis.

The interpretation considers the complete behaviour of the three components. The horizontal components are particularly important for evaluating long-term positional consistency, while the Up component generally presents greater scatter and is more sensitive to discontinuities.

4.2 Preliminary assessment criteria

Table 4 – Assessment Criteria

Continuity	Length and completeness of the daily coordinate series
Repeatability	Scatter of the North, East and Up residuals
Discontinuities	Offsets or anomalous intervals affecting the fitted trend
Velocity reliability	Magnitude and uncertainty of the estimated trend
Regional consistency	Agreement with neighbouring stations and the Eurasia-relative field

5. Preliminary Station-Stability Results

5.1 General behaviour

The majority of the QGEO-KZ stations provide multi-year coordinate series that can support an initial assessment of horizontal stability. The series nevertheless show different levels of scatter, data completeness and sensitivity to discontinuities. A detailed review of the individual plots is therefore required before a shortlist of potential IGS candidates is established.

The complete set of station plots is supplied separately as Annex A. It provides the evidence for the station-by-station assessment and allows the horizontal and vertical behaviour to be evaluated independently.

At this stage, the stations listed below require immediate review. A shortlist of the most promising stations will be prepared only after the complete set of coordinate time series has been assessed consistently.

5.2 Stations requiring immediate review

Table 5 – Stations with major problems

Station	Preliminary status	Observation
DKOM	Velocity/covariance solution not accepted	Review the coordinate series and identify the interval responsible for the unrealistic preliminary estimate.
HMYA	Velocity/covariance solution not accepted	Review the anomalous coordinate behaviour before attempting a new velocity estimate.
EGAN	Excluded	The four-character identifier duplicates an existing station name and the preliminary solution was not retained.
PARK	Excluded	The four-character identifier duplicates an existing station name and the preliminary solution was not retained.
LZHK	Flagged in the Eurasia-relative field	The current relative motion is inconsistent with the general network pattern and requires further analysis.
RBLS	Flagged in the Eurasia-relative field	The current relative motion is inconsistent with the general network pattern and requires further analysis.
XSHM	Flagged in the Eurasia-relative field	The current relative motion is inconsistent with the general network pattern and requires further analysis.

6. Motions Relative to the Eurasian Plate

The predicted rigid motion of the Eurasian plate was removed from the estimated horizontal velocities. The residual vectors provide an initial view of the internal consistency of the network and highlight stations whose behaviour differs from the regional pattern.

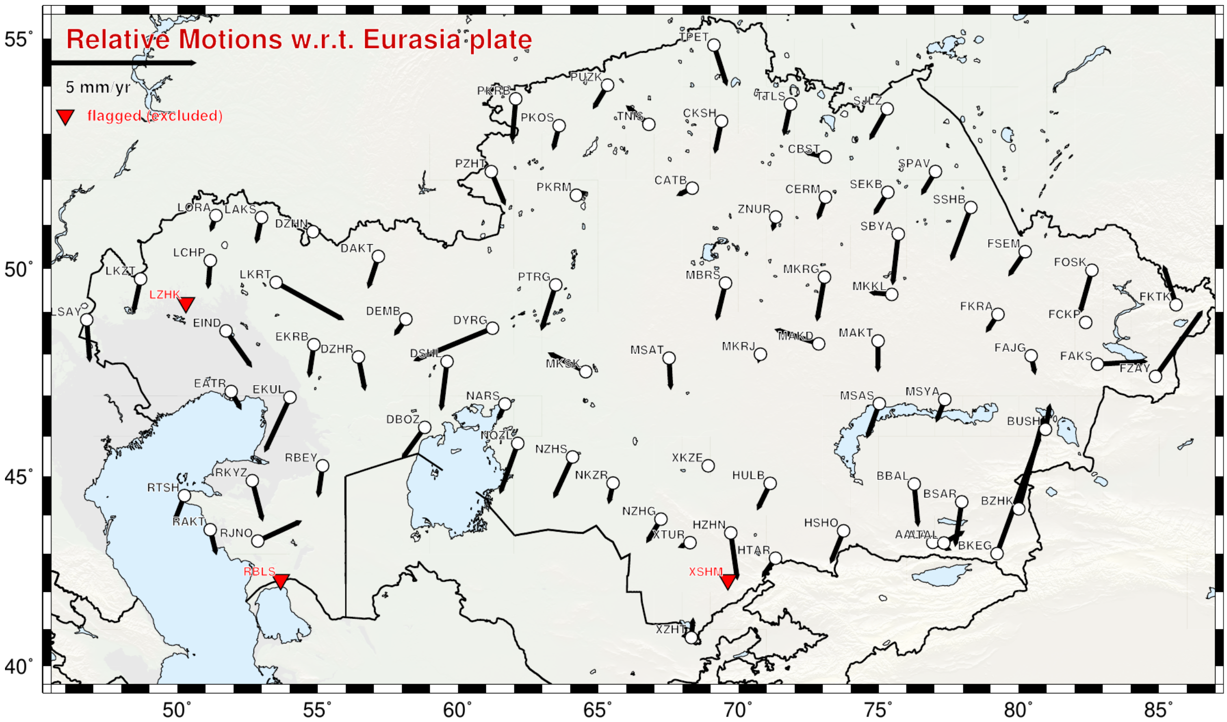

Figure 1. Preliminary horizontal velocities of the QGEO-KZ stations relative to the Eurasian plate. The reference vector is 5 mm/yr. Red symbols identify stations flagged and excluded from the current interpretation.

6.1 Preliminary interpretation

The Eurasia-relative field permits comparison of neighbouring stations after removal of the dominant rigid plate motion. Stations with coherent residual velocities provide greater confidence in the internal consistency of the network, while isolated or unusually large vectors require comparison with the corresponding coordinate time series.

The present field also provides a basis for evaluating possible long-term internal deformation. However, the relatively short observation span and the presence of unresolved processing discontinuities mean that no definitive deformation interpretation should yet be made.

7. Preliminary Conclusions

This first computation demonstrates that the QGEO-KZ observations can be used to construct multi-year daily coordinate time series and a regional velocity field. The analysis provides an initial basis for identifying the stations with the most promising long-term behaviour and for separating them from stations requiring further investigation.

The current results remain preliminary. The next assessment should address the identified discontinuities, review the anomalous station solutions and update the velocity estimates as additional observations become available. Only after this refinement should a shortlist be considered for a broader evaluation against the complete technical and operational expectations applicable to IGS stations.

Annex A - Complete Station Coordinate Time Series

The separate annex QGEO-KZ.pdf contains the North, East and Up coordinate time series for all analysed stations, together with the fitted trends and preliminary velocity estimates. These plots provide the principal evidence for the individual station-stability assessment.